F = ma in practice

LEARNING TARGET · R7 I can solve with F = ma — always finding the NET force first.

A · KEY VOCABULARY

Net force	Combine opposite forces: subtract and keep the direction of the larger one.
F = ma	Uses the *net* force, never just one pull.
Direction	Acceleration points the same way as the net force.
Balanced	Net force 0 N → a = 0 m/s² (still or constant velocity).

B · KEY EQUATION + UNITS

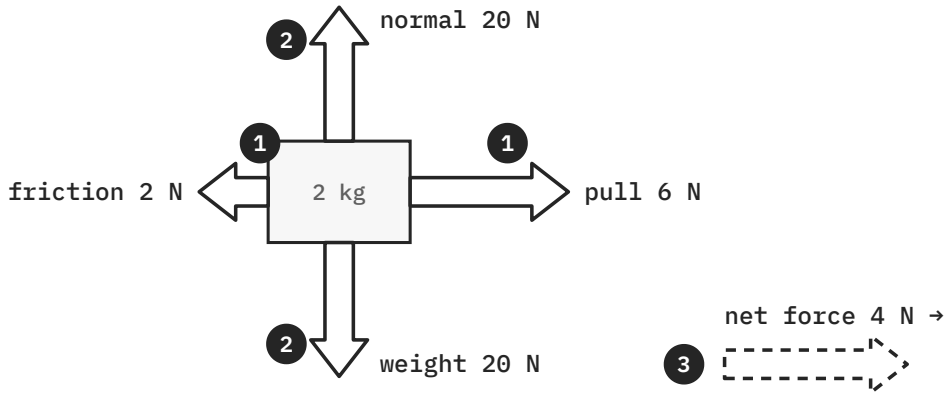
$$\Sigma F_x = m\ a_x$$

$$\Sigma F_x = F_{right} - F_{left}$$

F	newtons (N)
m · a	kg · m/s² (1 N = 1 kg·m/s²)

C · COLOR IT · THE ARROWS COMPETE

Vertical forces balance.



☐ **1 · ORANGE** Horizontal forces

☐ **2 · YELLOW** Vertical forces (balanced)

☐ **3 · ORANGE** Net force (thick arrow)

D · WORKED EXAMPLE

A 4 kg cart has 12 N pulling right and 4 N of friction left. Find the acceleration.

GIVEN	EQUATION	WORK	ANSWER
m = 4 kg 12 N right 4 N left	$a = F_{net} \div m$	$F_{net} = 12 - 4 = 8$ N right $a = 8 \div 4$	**2 m/s² right** with the net force

E · THE 2026-XJ THREAD

DAY 13 OF 22 · NEW TOOL: F = MA

F = ma links it all: from XJ's mass and the net force on it, we can calculate its acceleration, and from that, how its velocity will change.

Unit question: How fast is it moving, and where will it arrive?

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else